SANA SYED

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PROFESSIONAL SUMMARY

Second-year Computer Science (Artificial Intelligence) undergraduate with hands-on experience in machine learning, data analysis, and software development in Python, R, Java, and C#. Built an agentic AI fraud-detection platform (RAG, FastAPI, React) and a Random Forest fraud classifier (0.84 recall on highly imbalanced data). Seeking Software Engineering, Machine Learning, and AI Engineering internships.

EDUCATION

Bachelor of Computer Science (Hons) in Artificial Intelligence | Asia Pacific University of Technology & Innovation (APU), Kuala Lumpur, Malaysia
March 2025 – December 2027 (expected)
Key Modules: Programming for Data Analysis (R), Object-Oriented Development with Java, Introduction to OOP (C#), Database Systems, Introduction to AI, Mathematics for Computing, Integrated Computer Systems

TECHNICAL SKILLS

- **Languages:** Python, Java, C#, C++, R, SQL, NASM x86 Assembly
- **AI / Machine Learning:** Random Forest, Decision Trees, SMOTE, Hyperparameter Tuning (RandomizedSearchCV), ROC-AUC, Precision/Recall/F1, Agentic AI, RAG, Chain-of-Thought Prompting, Gemini API, scikit-learn
- **Data & Analysis:** Data Cleaning and Validation, Feature Engineering, Chi-Square Tests, ggplot2, Jupyter Notebook
- **Software & Databases:** FastAPI, React, Tailwind CSS, Java Swing, Windows Forms, MySQL, SQL Server, Firestore, OOP
- **Tools:** Git, GitHub, Docker, NetBeans, Visual Studio, Antigravity

PROJECTS

ScamSentinel MY – Real-Time Financial Fraud Detection | Project 2030: MyAI Future Hackathon, Track 5: Secure Digital | *Python, FastAPI, React, Gemini, Firestore*

- Built an agentic AI platform targeting digital scams behind more than RM1.3 billion in Malaysian losses in 2024, using Python 3.12, FastAPI, React 19, and Tailwind v4.
- Developed a Unified Threat Scanner covering 4 input channels (SMS messages, suspicious URLs, QR codes, and voice calls in beta) to detect phishing and social engineering in real time.
- Implemented a 5-step Chain-of-Thought risk scoring agent on Gemini 1.5 Flash that classifies the scam type, cross-references PDRM and BNM advisories through Vertex AI RAG over a 1,000+ document corpus, and returns a typed JSON risk verdict.
- Generated plain-language explanations of each verdict in English and Bahasa Malaysia.
- Created a Transaction Vault to check bank accounts, phone numbers, and e-wallet IDs against known mule-account records, plus an automated PDRM report generator.
- Built a decentralised Live Intel Feed using Firestore real-time listeners.

Credit Card Fraud Detection Using Random Forest | Group Research Paper (6 authors) | *Python, scikit-learn, imbalanced-learn*

- Built a Random Forest classifier on the 284,807-transaction Kaggle credit card dataset (0.17% fraud, 31 features), applying SMOTE to the training set only.
- Tuned 6 hyperparameters (number of trees, depth, split and leaf sizes, feature sampling, class weight) with RandomizedSearchCV and stratified cross-validation, sampling 20 candidates from a 96-combination grid because an exhaustive grid search was impractical on roughly 400,000 balanced training rows.
- Lowered the decision threshold to 0.3, raising fraud recall from 0.78 to 0.84 (124 of 148 frauds caught) on 85,443 test transactions at 99.93% accuracy, at the cost of false positives rising from 17 to 48.
- Reviewed existing work on Random Forest, XGBoost, SVM, logistic regression, autoencoders, and LSTMs, and identified a gap: few studies combine SMOTE with systematic hyperparameter tuning and recall-focused evaluation.

HR Employee Attrition Analysis and Prediction | Group Project (3 members) | *R, ggplot2*

- Cleaned a 2,000-record, 35-attribute HR dataset: removed 29 blank rows, standardised inconsistent labels, dropped 4 constant columns, and imputed missing values (median for numeric, mode for categorical).
- Enforced 5 validation rules (age 18 to 65, income above zero, years at company not above total working years, satisfaction scores 1 to 4, attrition coded 0 or 1), removing 163 impossible records to give a final 1,808-row dataset.
- Led the demographic analysis of attrition with distribution plots and Chi-Square tests: Age Group ($p = 1.4e-08$) and Marital Status ($p = 1.1e-06$) were significant, with employees aged 18 to 25 and single employees leaving most, while Gender was not significant ($p = 0.68$).

- Built a correlation heatmap, a Decision Tree, and a Random Forest (300 trees) on a stratified 70/30 split, handling the 84% to 16% class imbalance with ROC-AUC (Random Forest 0.717 vs Decision Tree 0.644), and added a risk scoring system.

APU-ASC Automotive Service Centre System | Group Project (3 members) | *Java, Swing*

- Developed the Technician module of a Java Swing desktop system serving 4 roles (Manager, Counter Staff, Technician, Customer): assigned appointments, job details, status updates, profile editing, and feedback.
- Implemented marking appointments as Completed and a feedback feature that sanitises comments (commas replaced with semicolons) to protect the integrity of the text-file records.
- Added a My Performance dashboard showing total, completed, and pending jobs and revenue, calculated through a TechnicianService class each time the dashboard loads.
- Worked within a three-tier architecture (UI, service, and file-handler layers) with plain-text storage, as databases were not permitted, and OOP design: abstract User class with 4 role subclasses, encapsulation, and a centralised Validator for input checks.

Internship Tracking and Management System (ITMS) | Group Project (3 members) | *C#, Windows Forms, SQL Server*

- Developed the Admin module of a 4-role desktop application (Admin, Company HR, Student, Supervisor) backed by a SQL Server LocalDB database, designed to replace spreadsheet and email-based internship tracking.
- Built 8 Admin screens: dashboard, student list, internship listings, student information editing, company list, uploaded reports, user data editing, and supervisor assignment.
- Authored the main storyboard, the Admin and Supervisor storyboards, code explanations, and test cases, plus the conclusion and workload matrix.

MediTrack – Pharmacy Management System | Individual Project | *NASM x86 Assembly, Linux*

- Built a menu-driven pharmacy management system in 32-bit NASM assembly with 4 modules: medicine management, inventory management, sales processing, and sales summary, controlled with conditional and unconditional jumps.
- Implemented array-based storage across 10 medicine slots with free-slot allocation, name search over fixed 20-byte records using a string comparison routine, low-stock checks, and restocking.
- Developed multi-item sales with stock reservation, quantity-based discounts, payment and change calculation, and stock updates after payment, reporting total sales, revenue, items sold, and discounts.
- Assembled with NASM and linked with GNU ld for 32-bit ELF under WSL2 Ubuntu, and documented the design with a flowchart, user manual, and annotated source code.
- Researched why assembly remains relevant to malware analysis, reverse engineering, and digital forensics.

Shape Generation System (Assembly vs Python) | Group Project (5 members) | *NASM Assembly, Python*

- Implemented the Heart routine of a menu-driven shape generator in NASM assembly (the other shapes were oval, square, rectangle, and triangle with size and fill options): printing a null-terminated ASCII art string from the data section, then waiting for Enter and returning to the main menu.
- Co-authored a report comparing the NASM program with an object-oriented Python version on development speed, readability, and I/O-bound performance, concluding that Python suits console output while assembly gives direct hardware control.
- Analysed how branch prediction and caching apply to the loops in the shape-drawing code.

CERTIFICATIONS, ACTIVITIES & LANGUAGES

Project 2030: MyAI Future Hackathon (Participation) | AI Tinkerers x Kiro: Agentic IDE Demo Day (Participation) | Docker: Container 101 | Public Speaking with Canva | Knowledge Transfer Program (KTP) Facilitator (Appreciation)
Languages: English (Full Professional) | Urdu (Full Professional) | Hindi (Professional Working) | Kashmiri (Native)